

Customer_Module_model_summary.txt

TBD2 pipeline output - rendered to PDF

NOTE ON THE SECTION BELOW: 'Overall Risk', the 'Risk Scenarios' ranking, and 'Follow-up Actions' are computed from randomly simulated stress-test scenarios (seeded for reproducibility), not from statistical evidence measured in this client's actual data. Treat them as illustrative stress-test projections, not as data-driven findings. (Model Evaluation Metrics further below ARE fit on the real data and are not affected by this note.)

Overall Risk: 0.3802845671230959

Risk Scenarios in Higher-First Priority:

Scenario_10: 0.9579666868632705

Scenario_9: 0.8037397818942802

Scenario_8: 0.638736195276424

Scenario_7: 0.48976899947419505

Scenario_6: 0.3611336711829887

Scenario_5: 0.25170614480786363

Scenario_4: 0.1591669199583243

Scenario_3: 0.09014483389627911

Scenario_2: 0.03943448241278718

Scenario_1: 0.011047955464545458

Average MSE across models is 9.79311356752109e-06, indicating strong confidence in the risk assessment.

Follow-up Actions:

- High risk in Scenario_10 (Avg Risk=0.958) driven by metrics: CHG10107, CRM10109, BIH10107. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.
- High risk in Scenario_9 (Avg Risk=0.804) driven by metrics: CFS10101, CBC10110, CSP10101. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.
- High risk in Scenario_8 (Avg Risk=0.639) driven by metrics: CFS10102, CFS10108, CIM10109. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.

Metrics involved in High-Risk Scenarios:

Scenario_10 (Avg Risk=0.958):

- CHG10107: Charge Reversal Rate
- CRM10109: Customer Engagement Rate
- BIH10107: Inquiry Accuracy Rate

Scenario_9 (Avg Risk=0.804):

- CFS10101: Response Rate
- CBC10110: Customer Satisfaction (CSAT) with Balance Management
- CSP10101: Segmentation Accuracy

Scenario_8 (Avg Risk=0.639):

- CFS10102: Customer Satisfaction (CSAT) Score
- CFS10108: Sentiment Analysis
- CIM10109: Data Usage Rate

Baseline Predictor (Mean of CUST_001):

MSE: 1.5311976057378582

R-squared: 0.0

RMSE: 1.2374156964164704

MAPE: 1.663849335374603%

PyTorch Model Evaluation:

MSE: 0.0009623768553458673

R-squared: 0.9993714874868276

RMSE: 0.031022199395688684

MAPE: 0.0403930972835058%

Model Evaluation Metrics:

MSE: 0.0009623768553458673

RMSE: 0.031022199395688684

R²: 0.9993714874868276

MAPE: 0.0403930972835058%

Risk Confidence Score: High

Recommendation - Continuously monitor performance metrics and adjust strategies as needed.

Recommendation - Reassess risk management plans periodically to ensure effectiveness.

-----End of Customer Module Risk Analysis Summary-----